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High previous-week training load amplifies hip adduction squeeze strength reductions in male academy football players with performance-limiting groin pain

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ABSTRACT

Objective: To examine whether preceding seven-day football training load and previous seven-day groin pain influence hip adduction squeeze strength in academy football players.
Design: Longitudinal observational study with twice weekly repeated measures over seven weeks.
Setting: West African academy football club.
Participants: Nineteen male academy football players (age 16.5 ± 0.9 [15-18] years).
Main outcome measures: Hip adduction squeeze strength, performance-limiting groin pain, and football training load were captured. Observations with and without groin pain were stratified into two groups, and two linear-mixed effects models examined hip adduction squeeze strength based on preceding-week football training load.
Results: Hip adduction squeeze strength was significantly influenced by football training load when associated with groin pain (estimate −0.13, 95% CI [-0.20, −0.07], $p < 0.001$). In observations without groin pain, football training load did not influence hip adduction squeeze strength (estimate −0.01, 95% CI [-0.06, 0.03], $p = 0.634$.
Conclusions: When groin pain is present, hip adduction squeeze strength is adversely influenced by higher football training loads. Players without groin pain do not experience hip adduction squeeze strength reductions from high football training load. Monitoring training load should be a consideration for practitioners working with football players that are experiencing groin pain.

1. Introduction

Groin pain is common in youth male football players (Andersen et al., 2023; Robles-Palazón et al., 2022), with most cases developing gradually due to overuse (Engebretsen et al., 2010; Mosler et al., 2018; Sonesson et al., 2023; Werner et al., 2009). As players mature during adolescence, increasing lower body mass requires greater force to accelerate the lower limb, thus increasing the biomechanical demand on the adductors (Dupré et al., 2020). Simultaneously, football exposure tends to increase rapidly from early adolescence (Light et al., 2021), increasing the risk of gradual-onset groin injuries. Delayed reporting between pain onset and medical staff being informed about pain may increase injury severity (Thorborg, Rathleff, et al., 2017), and players can often under-report symptoms (DeLang et al., 2021).

Previous groin injury, higher levels of play, and reduced hip adduction strength are known risk factors for groin injury in sport (Whittaker et al., 2015). Football players demonstrating weaker hip adduction strength are more likely to have groin problems compared to stronger players (Quintana-Cepedal et al., 2025). Thus, decreased hip adduction strength is the main modifiable risk factor for practitioners to consider. Measuring hip adduction strength may assist practitioners to reduce overuse groin injury risk (Moreno-Pérez et al., 2019). To capture groin pain before symptoms worsen, serial monitoring strategies have been suggested. Serial monitoring uses frequent adduction strength measures to capture changes and alert practitioners when players demonstrate strength reduction (Adams et al., 2024; DeLang et al., 2023; Wollin, Thorborg, et al., 2018). Decreases in hip adduction strength are associated with groin pain and have been observed in the week prior to reporting pain (Crow et al., 2010; DeLang et al., 2023). A prospective cohort study using strength-based alerts (exceeding minimal detectable

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change) reduced the burden of time-loss groin injury by 49% (Adams et al., 2024), supporting the use of serial monitoring for secondary prevention. Relying on objective hip adduction squeeze strength changes and subjective pain reporting may be sufficient to implement serial monitoring. However, individual variation in both painful and pain-free groups suggests other contextual factors may play a role in serial hip adduction squeeze strength (DeLang et al., 2023).

In professional football players, groin injury risk may be higher when accumulated chronic and acute training duration is also high (Mosler et al., 2022). Acute changes in adductor strength have been observed following football exposure, particularly after matches (Hamad et al., 2025; Springham et al., 2024). During a match-congested period at an international tournament, 70% of elite youth football players from one team experienced hip adduction squeeze strength reductions greater than 15%, despite only 45% reporting pain (Wollin, Pizzari, et al., 2018). The effects of football training load on serial hip adduction squeeze strength should be further considered in the context of implementing a serial monitoring system.

Previous studies have attempted to link training load and serial hip adduction squeeze strength using rating of perceived exertion (RPE). Esmaeili et al. (2018) reported trivial effects of RPE-based training load on hip adduction squeeze strength, while Tiernan et al. (2019) reported a weak negative correlation between training load (RPE * session duration) and hip adduction squeeze strength. Using an external load measure, such as Global Positioning Systems (GPS) data, may provide more insight into the effects of football training load on hip adduction squeeze strength.

To date, studies have examined the effects of pain and training load on hip adduction squeeze strength in isolation (Crow et al., 2010; DeLang et al., 2023; Esmaeili et al., 2018; Tiernan et al., 2019). However, isolating either factor may introduce confounding bias. Pain is well-established as a deterrent to hip adduction squeeze strength (Crow et al., 2010; DeLang et al., 2023; Nevin & Delahunt, 2014), so examining training load alone is not sufficient. On the contrary, studies examining pain have not sufficiently accounted for the week-to-week variation observed in pain-free players (DeLang et al., 2023), suggesting that training load may also be considered. The present study incorporates both pain and training load to examine serial hip adduction squeeze strength. The aim of this study is to examine whether preceding seven-day football training load and previous seven-day groin pain influence hip adduction squeeze strength in academy football players.

2. Materials and methods

Male players from one under-18 West African football academy were included in the study by convenience sampling. The academy participates in the highest level of football domestically and frequently participates in international tournaments. Players were included if participating in the under-18 squad at the academy and they did not have a time-loss injury at study initiation. A longitudinal observational study with repeated measures was conducted for seven weeks in-season during the summer of 2023. All study methods and procedures were approved by the University of Ghana College of Health Sciences Ethical and Protocol Review Committee and all participants signed consent. Participants under 18 had a parent/guardian sign on their behalf. The study was conducted in accordance with the ethical standards in the Declaration of Helsinki.

2.1. Outcomes & procedures

This study was conducted as a subset of an ongoing serial monitoring system to combat groin pain in academy football players. During the seven-week testing period, players were tested twice weekly (Monday and Thursday) using a serial monitoring protocol which included 1) subjective reports of groin pain during football over the previous seven days, 2) hip adduction squeeze strength testing, and 3) reported pain during the squeeze test. Each monitoring session occurred prior to the team training session on that day. Due to practical implementation of the study in an applied setting, no considerations were possible for effects of preceding 24-h fatigue. Groin pain was operationalized as "performance-limiting" even in the absence of time-loss (DeLang et al., 2022; Esteve et al., 2022). When athletes experienced a perceived deficit, their performance is negatively affected (Moritz et al., 2000). Groin pain over the preceding seven days was verbally asked at the start of each monitoring session with the question: "in the last week, have you had hip or groin pain that has limited your performance playing football?". If the participant responded yes, he was instructed to report the highest intensity groin pain experienced on a pain numeric rating scale (NRS) from 0 to 10 (0 = no pain, 10 = maximal imaginable pain). In the statistical models, groin pain was dichotomized, with NRS >0 signifying groin pain. For players with groin pain, time-loss was considered as an intervention strategy if necessary. Otherwise, normal standard of care for players with groin pain included exercise prescription modifications in the gym and/or pain-modulating strategies off-pitch.

Hip adduction squeeze strength was tested in a long-lever position using a valid and reliable portable groin strength device (GroinMate, CCAthletics, Odense, Denmark) (Ishøi et al., 2025). This testing device has consistent relative week-to-week reliability (intraclass coefficient 0.916-0.933) and absolute reliability (standard error of measurement percentage) of 7.6-8.6% (Ishøi et al., 2025). The testing procedure adds an objective component to the test setup for the Copenhagen 5-s squeeze test (5SST) (Thorborg et al., 2017). Players were instructed to perform a bilateral maximal effort isometric squeeze for 5 s with the cue "ready--go-1-2-3-4-5-and relax", with the first author (MDD) conducting all trials. One trial was performed at each monitoring session, which is sufficient to yield consistent results (Light & Thorborg, 2016). Each strength value is reported relative to the player's body mass (N/kg).

To examine football training load, global positioning systems (GPS) monitors were worn by all players during all on-pitch training sessions and matches (Catapult Global Positioning Systems, Australia). Studies of the Catapult system have indicated that practitioners can be confident in outputs from the system (Thornton et al., 2019), although there is increased variability in high-intensity actions (Johnston et al., 2014). Variables extracted from the GPS data included total distance (TD; m), high-speed running distance (HSD; m, >21 km/h), sprinting distance (SD; m, >25 km/h) accelerations and decelerations (ACCDEC; #; >2.5 m/s^2 and >-2.5 m/s^2), and on-pitch duration (DUR; minutes). GPS data for each player was then compiled into seven-day cumulative summaries preceding each serial monitoring observation and to coincide with the previous seven-day groin pain measure.

2.2. Statistical analyses

All statistical analyses were performed in R version 4.2.3. (R: The R Project for) A linear mixed-effects model was created in the *lme4* package (version 3.1.3), data visualizations used *sjPlot* (version 2.8.17) and *ggplot2*, and model analysis and assumption checks used *base*, *stats*, *performance* (version 0.12.4), and *car* (version 3.1.2).

The training load variables (TD, HSD, SD, ACCDEC, DUR) have high multicollinearity as assessed by variation inflation factors from a correlation plot (Supplementary File 1). A principal component analysis was conducted to reduce the related variables into independent principal components, where corresponding eigenvalues greater than one indicate a principal component accounts for more variance than any single original variable (Weaving et al., 2021). The first principal component had a proportion of variance of 81% and an eigenvalue of 4.06, while no other principal components had an eigenvalue greater than one (Supplementary File 1). Loading factors used to calculate total training load: TD*0.46, SD*0.40, HSR*0.45, DUR*0.45, ACCDEC*0.47. Training load used in the model was derived from the loading factors of the first principal component using the raw training load data grouped and scaled by individual players. Raw training load data is presented as

mean ± standard deviation in a descriptive table. Previous-week pain was dichotomized (0 = no pain, >0 = pain).

A linear mixed-effects model was used to examine hip adduction squeeze strength based on predictor variables previous-week groin pain, training load (a composite of each individual's data being grouped and scaled), the previous-week groin pain and training load interaction, and random intercepts for individual players (Supplementary File 2. By grouping and scaling individual's data, the models account for individual variations based on position and physical outputs. No assumptions were violated (diagnostic plots in Supplementary File 1).

Based on the positive interaction effect between groin pain and training load, the final models are stratified into two groups for ease of interpretation; 1) observations with groin pain and 2) observations without groin pain. Therefore, a linear-mixed effects model for observations with groin pain (ModelPain < - lmer (adduction ~ trainingload + (1 | name)) and observations without groin pain (ModelNoPain < - lmer (adduction ~ trainingload + (1 | name)) are presented.

3. Results

One player did not participate in the study due to a non-hip and groin-related time-loss injury when the study period began. Nineteen male academy football players (age 16.5 ± 0.9 [15-18] years, body mass 65.9 ± 7.6 kg, right-footed n = 13, left-footed n = 6) completed all monitoring sessions, comprising 266 total observations. Seventeen of 19 players reported groin pain during the study, with a total of 119 of 266 observations being associated with groin pain (44.7%). Although the majority of players experienced groin pain, no time-loss occurred for groin-related reasons during the study period. Descriptive data for each monitoring session, including the raw training load variables and adduction squeeze strength, are reported in Table 1.

In observations associated with groin pain, hip adduction squeeze strength decreased relative to higher football training loads (estimate = −0.13 N/kg, 95% CI [−0.20, −0.07], $p < 0.001$). In observations without presence of groin pain, hip adduction squeeze strength was not affected by football training load (estimate = −0.01 N/kg, 95% CI [−0.06, 0.03], p = 0.634) (Table 2, Fig. 1).

4. Discussion

This study examined whether preceding seven-day football training load and previous seven-day groin pain influences hip adduction squeeze strength in academy football players. Hip adduction squeeze strength decreases when higher football training loads occur – if the player is also experiencing groin pain. On the contrary, observations without previous-week groin pain show that football training load does not affect hip adduction squeeze strength.

Table 2
Results of linear mixed-effects models for the group with groin pain and the group without groin pain to examine whether training load influences hip adduction squeeze strength. SE, standard error; CI, confidence interval.

	Estimate	SE	95% CI	t-value	p-value
Groin Pain Group					
Intercept	3.14	0.19	[2.77, 3.52]	16.43	
Training Load	−0.13	0.04	[-0.20, −0.07]	−3.82	<0.001
No Groin Pain Group					
Intercept	3.37	0.15	[3.08, 3.66]	22.80	
Training Load	−0.01	0.02	[-0.06, 0.03]	−0.48	0.634

Previous studies have documented the relationship between current groin pain and adduction squeeze strength (Crow et al., 2010; DeLang et al., 2023; Nevin & Delahunt, 2014). In American academy players, players experiencing groin pain show decreased adduction squeeze strength that resolved when pain subsided (DeLang et al., 2023). Since decreased adduction strength is the main modifiable risk factor for groin injuries in sport (Quintana-Cepedal et al., 2025; Whittaker et al., 2015), non-time-loss groin problems associated with decreased strength may be detrimental to future groin health. The present study includes football training load, showing that when a player is already experiencing groin pain, he appears to be at an increased risk of further hip adduction strength degradation when exposed to high football training load.

For every one standard deviation increase in football training load, observations with groin pain were associated with −0.13 N/kg (−4%) weaker adduction strength. As decreased hip adduction strength is a known risk factor for groin injury (Quintana-Cepedal et al., 2025; Whittaker et al., 2015), the compounding effects of high football training load may lead to exacerbated symptoms. In professional football players, groin injury risk may be higher when accumulated acute and chronic training load are also high (Mosler et al., 2022). When groin pain is present, high football training load may further degrade hip adduction squeeze strength and contribute to persistent groin pain. Presumably, players experiencing multiple consecutive weeks with high football training load and groin pain may experience further strength deficits and increased time-loss injury risk.

Strength-based alerts during serial monitoring have been consistently observed in football players. In a monthly testing protocol, 40% of Australian academy players experienced decreases greater than 15%, and weekly testing of Australian professional players showed 46% experienced a decrease greater than minimal detectable change across an eight-point moving average (Adams et al., 2024). While pain is a known contributor to decreased adductor strength (Crow et al., 2010; DeLang et al., 2023), the present study may also provide insight into the additive effect of a high training load having an adverse effect on strength. Wollin et al. (Wollin, Pizzari, et al., 2018) observed 73% of

Table 1
Descriptive data for each monitoring session across all players, presented as mean ± standard deviation. Training load data is the sum of the preceding seven days. TD, total distance; HSR, high-speed running distance >20 km/h; SD, sprint distance >25 km/h; ACCDEC, accelerations + decelerations >2.5 m/s^2; DUR, duration; Strength, hip adduction squeeze strength.

Test Session	TD (m)	HSR (m)	SD (m)	ACCDEC (#)	DUR (min)	Strength (N/kg)
1	0 ± 0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	3.41 ± 0.74
2	13295 ± 1261	696.5 ± 212.6	59.1 ± 38.5	58.1 ± 21.5	161.7 ± 14.2	3.09 ± 0.78
3	27567 ± 1466	418.9 ± 254.0	86.0 ± 77.5	170.9 ± 34.7	354.7 ± 15.8	3.02 ± 0.75
4	32501 ± 1743	729.7 ± 285.3	148.1 ± 89.7	185.3 ± 38.5	378.9 ± 3.0	3.20 ± 0.71
5	33473 ± 4512	1150.7 ± 454.2	322.2 ± 170.3	177.6 ± 44.0	444.9 ± 63.3	3.20 ± 0.63
6	32779 ± 8926	1027.3 ± 530.0	306.7 ± 180.2	174.8 ± 65.0	439.5 ± 115.6	3.24 ± 0.69
7	32569 ± 8023	745.5 ± 318.8	165.7 ± 90.5	175.7 ± 59.0	435.8 ± 105.2	3.18 ± 0.81
8	31674 ± 7942	872.2 ± 402.1	166.9 ± 103.6	189.8 ± 67.8	450.4 ± 100.4	3.20 ± 0.67
9	31548 ± 8998	1152.0 ± 588.4	272.1 ± 155.5	223.6 ± 95.7	444.8 ± 129.3	3.09 ± 0.75
10	34628 ± 6104	1245.9 ± 569.5	349.8 ± 203.7	246.2 ± 89.9	470.8 ± 94.1	3.30 ± 0.74
11	28861 ± 7701	1094.8 ± 401.8	420.2 ± 164.1	194.7 ± 64.9	421.4 ± 106.9	3.21 ± 0.82
12	12466 ± 1672	536.4 ± 95.0	268.7 ± 49.0	100.9 ± 23.3	231.6 ± 22.3	3.42 ± 0.81
13	12212 ± 2928	450.8 ± 112.5	63.3 ± 35.6	103.2 ± 33.3	209.7 ± 52.3	3.26 ± 0.90
14	23486 ± 8026	792.4 ± 268.3	154.8 ± 96.6	163.5 ± 60.5	342.1 ± 103.3	3.29 ± 0.91

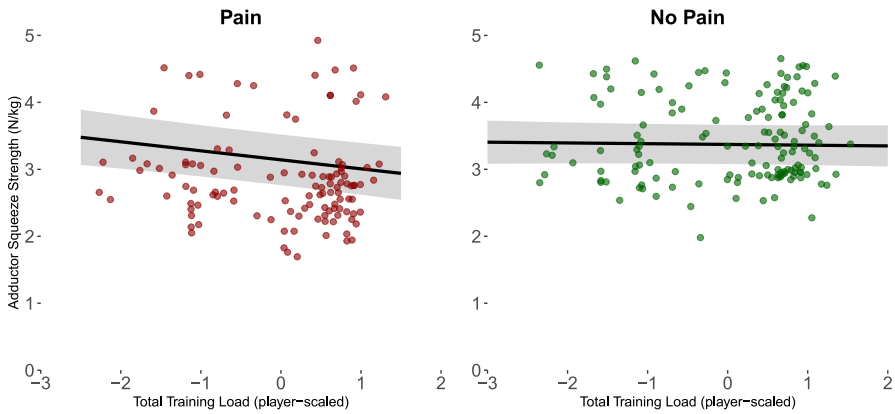

Fig. 1. Visual representations of the linear mixed-effects models (pain, left; no pain; right). Black lines represent predicted hip adduction squeeze strength values from the models. Red points represent the raw hip adduction squeeze strength values associated with pain; green points represent the raw hip adduction squeeze strength values associated with no pain. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

players experience decreases of greater than 15% during a period of match congestion at an academy tournament. In that study, cumulative rating of perceived exertion showed a negative relationship to short lever adduction squeeze strength (Wollin, Pizzari, et al., 2018). In the present study, football training load captured by GPS data showed that the presence of both groin pain and high football training loads may amplify strength deficits. Conversely, pain-free players appeared more resilient to cope with higher training loads and maintain adduction squeeze strength. In the context of serial monitoring, perhaps players with groin pain and high training loads may be more likely to be "flagged" for strength deficits.

The present findings may also give further insight into the management of non-time-loss groin pain cases captured through early detection. Previously, the role of pain modulation has been noted as a primary intervention strategy (DeLang et al., 2023). Pain cessation is related to restoring hip adduction squeeze strength (DeLang et al., 2023). Pain modulation strategies may include exercises (reciprocal inhibition, autogenic inhibition, etc.), manual therapy techniques, or activity modification. Given the present study's findings that pain and high training load combined are associated with greater hip adduction squeeze strength deficits, adjusting the intensifying stimulus (training load) may improve pain and assist with restoring hip adduction squeeze strength. In simpler terms – perhaps football training load modifications are important to resolve adduction strength deficits associated with groin pain.

4.1. Practical applications

Although no time-loss groin injuries occurred in the study, the association between performance-limiting groin pain, high training load, and decreased hip adduction squeeze strength is meaningful. Players with non-time-loss groin problems are more likely to experience future time-loss (Whalan et al., 2020), and decreased hip adduction squeeze strength is a known risk factor for time-loss groin injury (Quintana-Cepedal et al., 2025; Whittaker et al., 2015). Therefore, in the context of a serial monitoring system, a combined pain-load profile may give additional insights for early intervention on groin pain. When a player experiences groin pain and high previous-week training load, pain modulation strategies have been recommended to "break the cycle" and mitigate the risk of subsequent weeks demonstrating a continued downward trend in squeeze strength (DeLang et al., 2023). The present study can also add that one of the pain modulation strategies may be through modifying football activity (e.g., decreased football exposure).

Factoring training load into serial monitoring for groin pain appears to be relevant. Extrapolating the current findings might suggest that repeated weeks of high training load and performance-limiting groin pain would further reduce hip adduction squeeze strength, potentially leading to greater risk of time-loss injury. Particularly, the objective of serial monitoring is to capture these ongoing groin pain problems. Monitoring objective hip adduction squeeze strength, subjective reports of groin pain, and football training load may aid to combat eventual time-loss groin injuries in academy football players.

4.2. Limitations

This study is subjected to detection bias, as the structured approach to serial monitoring may capture subthreshold groin problems that may be otherwise unreported. However, operationally defining groin pain as "performance-limiting" suggests that subjectively, players perceive their symptoms as meaningful. Capturing groin pain early is an optimal strategy to mitigate the risks of time-loss and severe time-loss cases (Thorborg, 2023). A study over a longer period, also capturing time-loss cases, may further guide the how and when to intervene using secondary prevention. This study is limited by a small sample size (n = 19) and focus on a single elite West African academy cohort, meaning that future research should expand to larger, more diverse cohorts. Although all players in the sample are from the same team and participate together in training, the age range from 15 to 18 would suggest that from a maturational standpoint, not all players are alike. Previous studies have shown that adduction squeeze strength is consistent throughout academy years (e.g., adolescence) when presented relative to body mass (DeLang et al., 2020), as done in the present study. While the statistical analysis accounts for individual differences, the present study does not stratify training load results by playing position, which may be relevant to include in future studies. The present study shows evidence of the concept: pain and higher training loads compound into greater hip adduction squeeze strength deficits. However, it is not currently feasible to apply cut-off points or a specific algorithm at an individual level to prospectively identify at-risk players.

5. Conclusion

The presence of groin pain reducing hip adduction squeeze strength in football players has been previously reported. The present study shows that players experiencing groin pain are also more likely to show hip adduction squeeze strength deficits when undergoing high football training loads. Monitoring both pain and training load in tandem may enhance early detection and guide timely risk reduction strategies to mitigate risk factors for groin injury. When players are experiencing groin pain, practitioners may be cautious with ongoing high football

training loads to mitigate the risks of decreased hip adduction squeeze strength.

CRediT authorship contribution statement

Matthew D. DeLang: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Lasse Ishøi:** Writing – review & editing, Visualization, Methodology, Formal analysis, Conceptualization. **Kasper Thornton:** Writing – review & editing, Conceptualization. **Kristian Thorborg:** Writing – review & editing, Supervision, Project administration.

Ethical statement

All study methods and procedures were approved by the University of Ghana College of Health Sciences Ethical and Protocol Review Committee and all participants signed consent. Participants under 18 had a parent/guardian sign on their behalf. The study was conducted in accordance with the ethical standards in the Declaration of Helsinki.

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Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ptsp.2026.101902.

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